

Tangential integration and finite chart assembly of the stochastic Taylor tree (Programme S, tranche A3)

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Abstract

Units 278–285 carry the one-chart stochastic Taylor tree (Headline XXXV) through the two remaining steps of the paper’s §4.3 proof: integration over the tangential parameter of a stratum (the cited-but-unstated `lemma:AsymInt`) and the sum over finitely many charts. Tangential data are continuous maps $C(K, E_b \times E_b)$ from a compact parameter space into the weighted- ℓ^1 data space; the integrated integral and coefficients are Bochner integrals against a finite measure; the finite cutoff estimate is integrated (no infinite-series interchange), the integrated ordered remainders converge uniformly on sup-norm balls, and the probabilistic transfer of A2 applies verbatim (Headline XXXVI). For assembly, every chart integral is an abstract finite-cutoff expansion on its own lattice; expansions add, refine to a common lattice $Q^{-1}\mathbb{N}$ and pad to a common degree with the support theorems as justification; the global ordered remainder converges uniformly on joint data balls at every common-lattice target, including targets no single chart sees; and with an external chart decomposition $Z_\ell^0 = \sum_I \mathcal{Z}^I(N_\ell; X_{\ell,I}) + E_\ell$ whose residual is negligible at the target scale (in particular exponentially small in probability), the ordered normalised remainder of Z_ℓ^0 converges in distribution to the limiting global coefficient (Headline XXXVII). Reviews v29 and v30: pass. Eight units against a cap of eighteen. No `sorry`, no additional `axiom`; 3413 jobs; pin 5d5fb10.

1 The design (Astra #34)

Two layers for tangential integration: a finite-measure core (bounded measurable data) and the public interface $C(K, E)$ on a compact metrizable K , whose sup norm supplies the uniform ball control that the A2 estimates need; the partition-of-unity weight $\rho_I(v)$ absorbed into the amplitude family. For assembly: a dependent product over charts (not a sigma type, not padding of normal dimensions), a common rational lattice with zero coefficients where a chart’s own lattice or degree is exceeded, all charts expanded through one cutoff $L > \mu$, and the global predecessors subtracted after summation. Review v29 added the demand that the padding be a theorem (coefficients vanish off the local lattice and above the local degree) and that global targets not be handled by invoking local remainder theorems whose hypotheses fail.

2 Tangential integration (units 278–280)

`TangentialData K d = C(K, DataSpace d)` with the Borel σ -algebra; $\|x(v)\| \leq \|x\|$ (`ContinuousMap.norm_coe_le_norm`). Integrands are continuous on the compact K (composition of the ballwise-Lipschitz functionals of A1–A2 with the continuous map), hence integrable against a finite measure; the integrated cutoff estimate is the pointwise one integrated under a bound independent of v (`norm_integral_le_of_norm_le_const`, total mass $\nu(K)$). The integrated remainder

is the integral of the pointwise remainder (finite sums and division by the scalar $N^{-\mu}(\log N)^j$ pass through the Bochner integral), so the integrated majorant is $\nu(K)$ times the A2 majorant and convergence is uniform on sup-norm balls. The probability step is the A2 pipeline with the generic lemmas restated for an arbitrary Borel normed group (norm-boundedness in probability by port-manteau on $\{\|\cdot\| \geq m\}$, uniform-on-balls $\Rightarrow$ in probability, Slutsky), giving Headline XXXVI and its joint form.

3 Assembly (units 281–285)

Abstract expansions (u281). `CutoffExpansion Q D Z c`: for every cutoff $L > 0$ an eventual bound $KN^{-L}(1+\log N)^D$ on Z minus its spectral sum over $\Lambda_L^Q \times \{0..D\}$. Closed under sums; `refine` for $Q \mid Q'$ when the coefficients vanish off $Q^{-1}\mathbb{N}$ (equality of spectral sums by `Finset.sum_subset`); `pad` when they vanish above D . The abstract quantitative estimate: cutoff bound at $L = \mu + 1$ with constant K and coefficient bound C on the index set give $|R_N^{\mu,j} - c_{\mu,j}| \leq KN^{-(\mu+1)}(1 + \log N)^D/N^{-\mu} + C \sum_{\text{rest}} \text{termMajorant}$ for $N \geq e$ — the unit-275 argument with the ball bound replaced by C .

Charts as expansions (u282). Support theorems: `dataBoxCoeff` vanishes for $j > n$ (empty `Ico`), off the candidate set, hence off $(2 \prod k_i)^{-1}\mathbb{N}$; the chart cutoff bound in the sample size, $c^{-L}(1 + |\log c|)^n$ absorbing the box scale $c = b^{2|k|}$ (`one_add_log_mul_le`); every chart integral is a `CutoffExpansion`.

Global assembly (u283). `JointData K n = I, C(K_I, DataSpace (n_I+1))` as a `def` synonym carrying the sup norm and the *Borel* σ -algebra (the product σ -algebra need not coincide in a nonseparable function space); $Q = \prod_I Q_I$, $D = \max_I n_I$; global integral and coefficients are sums over charts. The global cutoff bound rewrites each chart’s common-lattice spectral sum to its own by the two support equalities and applies the chart bound; the global ordered remainder is then a direct instance of the abstract estimate with uniform constants on joint balls — so global targets off a chart’s lattice or above its degree need no separate argument, and predecessor compatibility is structural (the global remainder is defined from the common index set).

Stochastic assembly (u284–285). Continuous mapping on `JointData`, norm-boundedness in probability, uniform-on-balls, Slutsky give convergence of the assembled remainder; the external decomposition $Z_\ell^0 = \mathcal{Z}^{\text{glob}}(N_\ell; X_\ell) + E_\ell$ with $E_\ell/(N_\ell^{-\mu}(\log N_\ell)^j) \rightarrow 0$ in probability adds one more Slutsky step (the difference of the two normalised remainders is exactly the normalised residual, by `ring` under Lean’s totalised division). Unit 285 adds the bridge from the paper’s exponential residual ($E_\ell e^{\varepsilon N_\ell} \rightarrow 0$ in probability, via `tendsto_rpow_mul_exp_neg_mul_atTop_nhds_zero`) and the a.e. form of the decomposition (`TendstoInMeasure.congr`).

4 Reviews and non-claims

Review v29 (u278–280): pass, with the assembly prerequisites listed above. Review v30 (u281–284): pass, no blocking issue; nonblocking improvements done in u285. Non-claims: the function-space regularity and the joint convergence in distribution of the chart data are hypotheses (not derived from Hypothesis I, pointwise analyticity, or empirical-process theory); the chart decomposition, the partition of unity and its absorption into amplitudes, the away-from-minimum contribution and the identification with the original partition function are external; the common lattice is an indexing

superset of the paper’s Λ^* and chart contributions may cancel; no Gaussianity; deterministic sample sizes; targetwise statements (finite vectors of coefficients jointly).

5 Lessons

- **Abstract the bookkeeping once.** The finite-cutoff expansion abstraction turned chart assembly into three short lemmas (add, refine, pad) plus one quantitative estimate, and made “global targets a chart does not see” a non-issue.
- **Padding by theorem.** The support lemmas cost twenty lines and remove any doubt about what the common lattice means; the reviewer insisted, rightly.
- **Borel by declaration.** For products of function spaces, declare the Borel structure on a `def` synonym; do not rely on `Pi.borelSpace`, which needs second countability.
- **π is not a variable name.** With open `Real`, a named argument (`:= ...`) parses as the constant; Mathlib’s implicit family in `continuous_apply` is `A`.
- **Mechanise the gate.** All eight units passed `lake build` first or second time; the long-line linter caught something on five of them and is now a hard precondition of the commit.